TALENTRA DATABASE DESIGN HACKATHON

Milestone 4 — Completed Column Design Draft

B12–B15 | Profile and Jobs & Applications modules

Status: This version fills the column inventory for the entities discussed so far. It is a complete working draft for team review, not formal team approval.

Use PostgreSQL types and conventions below; reconcile all names and relationships with the approved Milestones 2–3 ER model.

1. Design conventions

• Singular snake_case names; surrogate `id` primary keys; foreign keys named `_id`.
• Business tables use `created_at`, `created_by`, `updated_at`, `updated_by`.
• `created_at` and `updated_at` are NOT NULL TIMESTAMPTZ; actor fields are nullable here for system-generated actions, subject to team approval.
• Store instants in UTC; use NUMERIC for money plus a currency code; use lookup tables for controlled types/statuses.
• Add real FK, CHECK, UNIQUE and deletion rules during implementation.

2. Profile module — column inventory

candidate_profile

Column	Type	Null?	Rule / purpose
id	BIGINT	No	PK.
user_id	BIGINT	No	FK to user/account; UNIQUE if one profile per user.
first_name	VARCHAR(150)	No	Personal data.
last_name	VARCHAR(150)	Yes	Personal data.
date_of_birth	DATE	Yes	Sensitive personal data; collect only if needed.
headline	VARCHAR(250)	Yes	Professional headline.
summary	TEXT	Yes	Visibility controlled by profile settings.
country_id	BIGINT	Yes	FK to country/location lookup.
city_id	BIGINT	Yes	FK to city/location lookup.
preferred_language_id	BIGINT	Yes	FK to language lookup.
created_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
created_by	BIGINT	Yes	Actor/user ID; nullable for system actions.
updated_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
updated_by	BIGINT	Yes	Actor/user ID; nullable for system actions.
deleted_at	TIMESTAMPTZ	Yes	Soft-delete marker; NULL means active.

candidate_education

Column	Type	Null?	Rule / purpose
id	BIGINT	No	PK.
candidate_profile_id	BIGINT	No	FK to candidate_profile.
qualification_id	BIGINT	No	FK to qualification lookup.
institution_name	VARCHAR(200)	No	Institution name.
field_of_study	VARCHAR(200)	Yes	Specialization.
start_date	DATE	Yes	Check end_date >= start_date when both exist.
end_date	DATE	Yes	Nullable for ongoing/unknown study.
grade	VARCHAR(100)	Yes	Keep flexible across grading systems.
created_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
created_by	BIGINT	Yes	Actor/user ID; nullable for system actions.
updated_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
updated_by	BIGINT	Yes	Actor/user ID; nullable for system actions.

candidate_work_experience

Column	Type	Null?	Rule / purpose
id	BIGINT	No	PK.
candidate_profile_id	BIGINT	No	FK to candidate_profile.
job_role_id	BIGINT	Yes	FK to job role.
company_name	VARCHAR(200)	No	Employer.
employment_type_id	BIGINT	Yes	FK to employment type lookup.
start_date	DATE	Yes	Check date order.
end_date	DATE	Yes	NULL if current/unknown.
is_current	BOOLEAN	No	DEFAULT FALSE; if true, end_date should be NULL.
description	TEXT	Yes	Role details.
created_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
created_by	BIGINT	Yes	Actor/user ID; nullable for system actions.

Column	Type	Null?	Rule / purpose
updated_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
updated_by	BIGINT	Yes	Actor/user ID; nullable for system actions.

candidate_project

Column	Type	Null?	Rule / purpose
id	BIGINT	No	PK.
candidate_profile_id	BIGINT	No	FK to candidate_profile.
project_name	VARCHAR(200)	No	Project name.
role	VARCHAR(150)	Yes	Candidate role.
start_date	DATE	Yes	Check date order.
end_date	DATE	Yes	Optional end date.
outcome	TEXT	Yes	Project outcome.
project_url	TEXT	Yes	Optional URL; validate format.
created_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
created_by	BIGINT	Yes	Actor/user ID; nullable for system actions.
updated_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
updated_by	BIGINT	Yes	Actor/user ID; nullable for system actions.

candidate_skill

Column	Type	Null?	Rule / purpose
id	BIGINT	No	PK.
candidate_profile_id	BIGINT	No	FK to candidate_profile.
skill_id	BIGINT	No	FK to skill lookup.
proficiency_level_id	BIGINT	Yes	FK to proficiency lookup.
created_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
created_by	BIGINT	Yes	Actor/user ID; nullable for system actions.
updated_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
updated_by	BIGINT	Yes	Actor/user ID; nullable for system actions.

candidate_language

Column	Type	Null?	Rule / purpose
id	BIGINT	No	PK.
candidate_profile_id	BIGINT	No	FK to candidate_profile.
language_id	BIGINT	No	FK to language lookup.
proficiency_level_id	BIGINT	Yes	FK to proficiency lookup.
created_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
created_by	BIGINT	Yes	Actor/user ID; nullable for system actions.
updated_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
updated_by	BIGINT	Yes	Actor/user ID; nullable for system actions.

2.1 Resume and candidate development

resume

Column	Type	Null?	Rule / purpose
id	BIGINT	No	PK.
candidate_profile_id	BIGINT	No	FK to candidate_profile.
title	VARCHAR(200)	No	Resume title.
created_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
created_by	BIGINT	Yes	Actor/user ID; nullable for system actions.
updated_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
updated_by	BIGINT	Yes	Actor/user ID; nullable for system actions.

resume_version

Column	Type	Null?	Rule / purpose
id	BIGINT	No	PK.
resume_id	BIGINT	No	FK to resume.
version_number	INTEGER	No	CHECK > 0; UNIQUE(resume_id, version_number).
resume_content	TEXT	No	Resume content; restrict access.
language_code	VARCHAR(10)	Yes	Standardize language tag.
resume_template_id	BIGINT	Yes	FK to resume_template.
created_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
created_by	BIGINT	Yes	Actor/user ID; nullable for system actions.
updated_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
updated_by	BIGINT	Yes	Actor/user ID; nullable for system actions.

resume_template

Column	Type	Null?	Rule / purpose
id	BIGINT	No	PK.
template_name	VARCHAR(150)	No	Consider UNIQUE.
template_description	TEXT	Yes	Optional description.
created_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
created_by	BIGINT	Yes	Actor/user ID; nullable for system actions.
updated_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
updated_by	BIGINT	Yes	Actor/user ID; nullable for system actions.

candidate_assessment

Column	Type	Null?	Rule / purpose
id	BIGINT	No	PK.
candidate_profile_id	BIGINT	No	FK to candidate_profile.
assessment_type_id	BIGINT	No	FK to assessment type.
score	NUMERIC(6,2)	Yes	Validate against assessment's scoring scale.
attempted_at	TIMESTAMPTZ	Yes	UTC attempt time.
created_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
created_by	BIGINT	Yes	Actor/user ID; nullable for system actions.
updated_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
updated_by	BIGINT	Yes	Actor/user ID; nullable for system actions.

course_enrolment

Column	Type	Null?	Rule / purpose
id	BIGINT	No	PK.
candidate_profile_id	BIGINT	No	FK to candidate_profile.
course_name	VARCHAR(200)	No	Course title.
enrolled_at	TIMESTAMPTZ	Yes	UTC time.
completed_at	TIMESTAMPTZ	Yes	Must not precede enrolled_at.

Column	Type	Null?	Rule / purpose
created_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
created_by	BIGINT	Yes	Actor/user ID; nullable for system actions.
updated_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
updated_by	BIGINT	Yes	Actor/user ID; nullable for system actions.

candidate_contact

Column	Type	Null?	Rule / purpose
id	BIGINT	No	PK.
candidate_profile_id	BIGINT	No	FK to candidate_profile.
contact_type_id	BIGINT	No	FK to contact_type lookup.
contact_value	VARCHAR(320)	No	Email/phone/value; restrict access.
is_primary	BOOLEAN	No	DEFAULT FALSE.
is_verified	BOOLEAN	No	DEFAULT FALSE.
verified_at	TIMESTAMPTZ	Yes	Set when verification occurs.
visibility	VARCHAR(20)	No	Prefer lookup or CHECK of approved values.
created_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
created_by	BIGINT	Yes	Actor/user ID; nullable for system actions.
updated_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
updated_by	BIGINT	Yes	Actor/user ID; nullable for system actions.

profile_completion_score

Column	Type	Null?	Rule / purpose
id	BIGINT	No	PK.
candidate_profile_id	BIGINT	No	FK; UNIQUE if only latest score.
score	NUMERIC(5,2)	No	CHECK BETWEEN 0 AND 100.
calculated_at	TIMESTAMPTZ	No	Calculation time.
calculation_version	VARCHAR(30)	No	Scoring logic version.
created_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
created_by	BIGINT	Yes	Actor/user ID; nullable for system actions.
updated_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
updated_by	BIGINT	Yes	Actor/user ID; nullable for system actions.

2.2 Profile activity, search, shortlist, bookmark and referral

candidate_activity_log

Column	Type	Null?	Rule / purpose
id	BIGINT	No	PK.
candidate_profile_id	BIGINT	No	FK to candidate_profile.
activity_type_id	BIGINT	No	FK to activity type lookup.
occurred_at	TIMESTAMPTZ	No	UTC event time.
metadata	JSONB	Yes	Optional structured metadata; avoid secrets/unnecessary personal data.
actor_user_id	BIGINT	Yes	FK to user/actor.
created_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
created_by	BIGINT	Yes	Actor/user ID; nullable for system actions.
updated_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
updated_by	BIGINT	Yes	Actor/user ID; nullable for system actions.

candidate_profile_view

Column	Type	Null?	Rule / purpose
id	BIGINT	No	PK.
candidate_profile_id	BIGINT	No	Viewed profile FK.
viewer_user_id	BIGINT	Yes	Viewer FK; nullable for anonymous view if allowed.
viewed_at	TIMESTAMPTZ	No	UTC view time.
source	VARCHAR(50)	Yes	Optional source/channel.
created_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
created_by	BIGINT	Yes	Actor/user ID; nullable for system actions.
updated_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
updated_by	BIGINT	Yes	Actor/user ID; nullable for system actions.

candidate_job_search_history

Column	Type	Null?	Rule / purpose
id	BIGINT	No	PK.
candidate_profile_id	BIGINT	No	FK to candidate_profile.
search_query	TEXT	Yes	May contain personal data; limit retention.
filters	JSONB	Yes	Validate keys and supported filters.
searched_at	TIMESTAMPTZ	No	UTC search time.
created_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
created_by	BIGINT	Yes	Actor/user ID; nullable for system actions.
updated_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
updated_by	BIGINT	Yes	Actor/user ID; nullable for system actions.

candidate_application_history

Column	Type	Null?	Rule / purpose
id	BIGINT	No	PK.
candidate_profile_id	BIGINT	No	FK to candidate_profile.
job_application_id	BIGINT	No	FK to job_application.
event_type_id	BIGINT	No	FK to event/status lookup.
event_at	TIMESTAMPTZ	No	UTC event time.
details	JSONB	Yes	Optional; avoid duplicating status history.
created_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
created_by	BIGINT	Yes	Actor/user ID; nullable for system actions.
updated_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
updated_by	BIGINT	Yes	Actor/user ID; nullable for system actions.

candidate_shortlist

Column	Type	Null?	Rule / purpose
id	BIGINT	No	PK.
candidate_profile_id	BIGINT	No	FK to candidate_profile.
organisation_id	BIGINT	No	FK to organisation.
shortlisted_by	BIGINT	No	FK to recruiter/user.
shortlisted_at	TIMESTAMPTZ	No	UTC time.
notes	TEXT	Yes	Restrict access.
created_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
created_by	BIGINT	Yes	Actor/user ID; nullable for system actions.
updated_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
updated_by	BIGINT	Yes	Actor/user ID; nullable for system actions.

candidate_bookmark

Column	Type	Null?	Rule / purpose
id	BIGINT	No	PK.
candidate_profile_id	BIGINT	No	FK to candidate_profile.
job_requirement_id	BIGINT	No	FK to job_requirement; this draft supports job bookmarks.
bookmarked_at	TIMESTAMPTZ	No	UTC time.
label	VARCHAR(100)	Yes	Optional user label.
created_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
created_by	BIGINT	Yes	Actor/user ID; nullable for system actions.
updated_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
updated_by	BIGINT	Yes	Actor/user ID; nullable for system actions.

referral

Column	Type	Null?	Rule / purpose
id	BIGINT	No	PK.
referrer_user_id	BIGINT	No	FK to referring user.
referred_candidate_profile_id	BIGINT	Yes	FK to candidate profile; nullable until registration if allowed.
job_requirement_id	BIGINT	Yes	FK to related job.
referral_code	VARCHAR(80)	No	UNIQUE generated code.
status_id	BIGINT	No	FK to referral status lookup.
referred_at	TIMESTAMPTZ	No	UTC referral time.
converted_at	TIMESTAMPTZ	Yes	Must not precede referred_at.
created_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
created_by	BIGINT	Yes	Actor/user ID; nullable for system actions.
updated_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
updated_by	BIGINT	Yes	Actor/user ID; nullable for system actions.

3. Jobs & Applications — column inventory

job_requirement

Column	Type	Null?	Rule / purpose
id	BIGINT	No	PK.
organisation_id	BIGINT	No	FK to organisation.
title	VARCHAR(200)	No	Job title.
description	TEXT	No	Job description.
job_role_id	BIGINT	No	FK to job role.
employment_type_id	BIGINT	No	FK to employment type.
work_type_id	BIGINT	No	FK to work type.
status_id	BIGINT	No	FK to job status.
is_published	BOOLEAN	No	DEFAULT FALSE.
open_to_consultants	BOOLEAN	No	DEFAULT FALSE.
minimum_salary	NUMERIC(12,2)	Yes	Nonnegative lower bound.
maximum_salary	NUMERIC(12,2)	Yes	Nonnegative; >= minimum when both supplied.
currency_code	CHAR(3)	Yes	Required if either salary amount is set.
published_at	TIMESTAMPTZ	Yes	UTC publication time.
closes_at	TIMESTAMPTZ	Yes	Must follow published_at if both exist.
deleted_at	TIMESTAMPTZ	Yes	Soft-delete marker.
created_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
created_by	BIGINT	Yes	Actor/user ID; nullable for system actions.
updated_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
updated_by	BIGINT	Yes	Actor/user ID; nullable for system actions.

job_position

Column	Type	Null?	Rule / purpose
id	BIGINT	No	PK.
job_requirement_id	BIGINT	No	FK to job_requirement.
position_title	VARCHAR(200)	No	Position title.
vacancy_count	INTEGER	No	CHECK > 0.
location_id	BIGINT	Yes	FK to location.
created_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
created_by	BIGINT	Yes	Actor/user ID; nullable for system actions.
updated_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
updated_by	BIGINT	Yes	Actor/user ID; nullable for system actions.

job_questionnaire

Column	Type	Null?	Rule / purpose
id	BIGINT	No	PK.
job_requirement_id	BIGINT	No	FK to job_requirement.
title	VARCHAR(200)	No	Questionnaire title.
instructions	TEXT	Yes	Optional instructions.
is_active	BOOLEAN	No	DEFAULT TRUE.
created_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
created_by	BIGINT	Yes	Actor/user ID; nullable for system actions.
updated_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
updated_by	BIGINT	Yes	Actor/user ID; nullable for system actions.

questionnaire_question

Column	Type	Null?	Rule / purpose
id	BIGINT	No	PK.
job_questionnaire_id	BIGINT	No	FK to job_questionnaire.
question_text	TEXT	No	Question prompt.

Column	Type	Null?	Rule / purpose
question_type_id	BIGINT	No	FK to question type.
is_required	BOOLEAN	No	DEFAULT FALSE.
display_order	INTEGER	No	CHECK > 0.
options	JSONB	Yes	For choice types only; validate structure.
created_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
created_by	BIGINT	Yes	Actor/user ID; nullable for system actions.
updated_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
updated_by	BIGINT	Yes	Actor/user ID; nullable for system actions.

job_application

Column	Type	Null?	Rule / purpose
id	BIGINT	No	PK.
candidate_profile_id	BIGINT	No	FK to candidate_profile.
job_requirement_id	BIGINT	No	FK to job_requirement.
status_id	BIGINT	No	FK to application status.
applied_at	TIMESTAMPTZ	No	UTC application time.
cover_letter	TEXT	Yes	Optional cover letter.
resume_version_id	BIGINT	Yes	FK to resume_version used for application.
created_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
created_by	BIGINT	Yes	Actor/user ID; nullable for system actions.
updated_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
updated_by	BIGINT	Yes	Actor/user ID; nullable for system actions.

3.1 Application lifecycle and history entities

application_status_history

Column	Type	Null?	Rule / purpose
id	BIGINT	No	PK.
job_application_id	BIGINT	No	FK to job_application.
previous_status_id	BIGINT	Yes	NULL for first status.
new_status_id	BIGINT	No	FK to application status.
changed_at	TIMESTAMPTZ	No	UTC transition time.
changed_by	BIGINT	Yes	FK to user/actor.
reason	TEXT	Yes	Optional reason.
created_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
created_by	BIGINT	Yes	Actor/user ID; nullable for system actions.
updated_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
updated_by	BIGINT	Yes	Actor/user ID; nullable for system actions.

application_questionnaire_response

Column	Type	Null?	Rule / purpose
id	BIGINT	No	PK.
job_application_id	BIGINT	No	FK to job_application.
questionnaire_question_id	BIGINT	No	FK to questionnaire_question.
response_text	TEXT	Yes	For typed responses, consider JSONB or normalized answer values.
created_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
created_by	BIGINT	Yes	Actor/user ID; nullable for system actions.
updated_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
updated_by	BIGINT	Yes	Actor/user ID; nullable for system actions.

application_shortlist

Column	Type	Null?	Rule / purpose
id	BIGINT	No	PK.
job_application_id	BIGINT	No	FK to job_application.
shortlisted_by	BIGINT	No	FK to recruiter/user.
shortlisted_at	TIMESTAMPTZ	No	UTC time.
notes	TEXT	Yes	Recruiter-only notes.
created_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
created_by	BIGINT	Yes	Actor/user ID; nullable for system actions.
updated_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
updated_by	BIGINT	Yes	Actor/user ID; nullable for system actions.

interview

Column	Type	Null?	Rule / purpose
id	BIGINT	No	PK.
job_application_id	BIGINT	No	FK to job_application.
scheduled_at	TIMESTAMPTZ	No	UTC scheduled time.
interview_mode_id	BIGINT	No	FK to interview mode.
status_id	BIGINT	No	FK to interview status.
feedback	TEXT	Yes	Restrict access.
created_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
created_by	BIGINT	Yes	Actor/user ID; nullable for system actions.
updated_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
updated_by	BIGINT	Yes	Actor/user ID; nullable for system actions.

job_offer

Column	Type	Null?	Rule / purpose
id	BIGINT	No	PK.
job_application_id	BIGINT	No	FK to job_application.
offered_salary	NUMERIC(12,2)	No	CHECK >= 0.
currency_code	CHAR(3)	No	Currency code.
offered_at	TIMESTAMPTZ	No	UTC offer time.
expires_at	TIMESTAMPTZ	Yes	Exact expiry instant; must be after offered_at.
status_id	BIGINT	No	FK to offer status.
created_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
created_by	BIGINT	Yes	Actor/user ID; nullable for system actions.
updated_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
updated_by	BIGINT	Yes	Actor/user ID; nullable for system actions.

application_joining

Column	Type	Null?	Rule / purpose
id	BIGINT	No	PK.
job_application_id	BIGINT	No	FK; UNIQUE if one joining record per application.
joining_date	DATE	Yes	Clarify expected vs actual date.
joining_status_id	BIGINT	No	FK to joining status.
created_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
created_by	BIGINT	Yes	Actor/user ID; nullable for system actions.
updated_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
updated_by	BIGINT	Yes	Actor/user ID; nullable for system actions.

job_search_history

Column	Type	Null?	Rule / purpose
id	BIGINT	No	PK.
candidate_profile_id	BIGINT	No	FK to candidate_profile.
search_query	TEXT	Yes	Optional search text.
filters	JSONB	Yes	Validated search filters.
searched_at	TIMESTAMPTZ	No	UTC search time.
created_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
created_by	BIGINT	Yes	Actor/user ID; nullable for system actions.
updated_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
updated_by	BIGINT	Yes	Actor/user ID; nullable for system actions.

job_view_history

Column	Type	Null?	Rule / purpose
id	BIGINT	No	PK.
job_requirement_id	BIGINT	No	FK to job_requirement.
viewer_user_id	BIGINT	Yes	FK to user; nullable for anonymous view if allowed.
viewed_at	TIMESTAMPTZ	No	UTC view time.
source	VARCHAR(50)	Yes	Optional channel.
created_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
created_by	BIGINT	Yes	Actor/user ID; nullable for system actions.
updated_at	TIMESTAMPTZ	No	Audit timestamp, UTC.
updated_by	BIGINT	Yes	Actor/user ID; nullable for system actions.

4. Proposed constraints and indexes

Table(s)	Proposed rule	Reason
candidate_education / work / project	CHECK end_date IS NULL OR start_date IS NULL OR end_date >= start_date; for work, CHECK NOT is_current OR end_date IS NULL.	Valid date sequences.
candidate_skill	UNIQUE(candidate_profile_id, skill_id).	Prevent duplicate skill assignments.
candidate_language	UNIQUE(candidate_profile_id, language_id).	Prevent duplicate language entries.
resume_version	UNIQUE(resume_id, version_number); CHECK(version_number > 0).	Unique positive versions.
profile_completion_score	CHECK(score BETWEEN 0 AND 100); index candidate_profile_id, calculated_at DESC.	Valid score and latest-score access.
candidate_contact	Optional partial unique index for one primary contact per type/profile; approved visibility values.	Consistent primary contact and access values.
candidate_bookmark	UNIQUE(candidate_profile_id, job_requirement_id).	Prevent duplicate job bookmarks.
referral	UNIQUE(referral_code); CHECK(converted_at IS NULL OR converted_at >= referred_at).	Unique code and valid timeline.
job_requirement	Nonnegative salary checks; max >= min when both set; currency required when salary exists; closes_at > published_at if both exist.	Valid job and salary data.
job_position	CHECK(vacancy_count > 0).	At least one vacancy.
questionnaire_question	CHECK(display_order > 0).	Positive ordering.
job_application	UNIQUE(candidate_profile_id, job_requirement_id) only if repeat applications are forbidden.	Business rule must be confirmed.
application_status_history	Index(job_application_id, changed_at DESC); FK status and actor references.	Fast, auditable history.
job_offer	CHECK(offered_salary >= 0); CHECK(expires_at IS NULL OR expires_at > offered_at).	Valid offer.
application_joining	Optional UNIQUE(job_application_id) if one joining record is allowed.	Enforce confirmed cardinality.
Search/view/activity history	Indexes on parent IDs and event timestamps; define retention policy.	Performance and privacy.

Index based on actual query patterns; every extra index increases write and storage cost.

5. Data classification and lifecycle

• Personal: candidate name, date of birth, contact details, resume content, profile/search/view/activity history. Limit access and define retention.
• Recruitment-sensitive: assessment scores, questionnaire responses, shortlist notes, interview feedback, and offers. Restrict to authorized roles.
• Potentially public: headline, summary, skills, education, experience and projects only when candidate visibility settings allow.
• Soft delete is proposed for candidate_profile and job_requirement; define child-table behavior and history retention separately.
• Avoid secrets and unnecessary personal data in JSONB metadata. Decide whether historical records must be anonymized after account deletion.

6. B15 validation checklist

■ All entities have a proposed column inventory, including the previously pending tables.

■ Every entity has an id PK; all FK columns reference approved entities.

■ Audit fields are consistent and timestamp values are UTC-aware.

■ Every nullable column has a reason; personal data collection is justified.

■ CHECK, UNIQUE, FK and NOT NULL rules encode agreed business rules.

■ Controlled status/type/proficiency values use lookup tables.

■ Salary, date range, positive vacancy count, score, and resume-version rules are covered.

■ History tables have distinct purposes and do not conflict with canonical state.

■ Deletion, retention, access and anonymization policies are agreed.

■ Names and relationships match the approved Milestones 2–3 ER model.

7. Decisions required for team sign-off

1. Is one candidate profile allowed per user? If not, remove the proposed uniqueness rule.
2. Can a candidate reapply to the same job? Decide whether to enforce a unique candidate/job pair.
3. Is candidate_application_history distinct from application_status_history? Remove it if redundant or define its separate purpose.
4. Does candidate_bookmark support only jobs? This draft assumes job bookmarks.
5. Does profile_completion_score store every calculation or only the latest score?
6. Is application_joining one record per application, and does joining_date mean expected or actual?
7. Confirm exact lookup entities: user, organisation, location, job_role, qualification, language, employment_type, status types, proficiency, contact_type, activity_type, assessment_type, interview_mode, and related lookups.
8. Confirm actor nullability, audit rules, deletion policy, and retention rules.

8. Presentation guide

Suggested explanation: “Milestone 4 translates our entity-relationship design into a PostgreSQL data dictionary. For each entity, we define column names, data types, nullability, foreign keys, and audit fields. We use TIMESTAMPTZ for event times, NUMERIC plus currency code for money, and lookup tables for controlled statuses and types. We also propose checks for salary ranges, date ranges, vacancy counts, scores, and resume versions. We have documented the remaining business decisions so the team can approve a consistent schema before implementation.”

Panel follow-ups: TIMESTAMPTZ makes event instants unambiguous; NUMERIC avoids floating-point rounding for money; lookup tables enforce consistent controlled values; history tables preserve status transitions; JSONB is useful for variable filters but needs schema validation and privacy controls.

Completion: The column inventory is now drafted for all entities listed in this document. Treat Milestone 4 as approved only after the team resolves the decisions above and verifies the design against the hackathon docket and approved ER model.